THE BRAUER GROUP OF AN ALGEBRAICALLY CLOSED FIELD IS TRIVIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the elementwise reduction is carried out for an arbitrary element and the two inputs are named, so no step is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Finite central skew field extensions. Same chapter, the description of the Brauer group as consisting of classes of finite central skew field extensions of the ground field. Supplies that it suffices to show that a finite central skew field extension of an algebraically closed field is the field itself.
- Finite integral subalgebras of a field. Algebra, Lemma algebra-lemma-integral-over-field. Supplies that the subalgebra generated by one element of a skew field is a commutative finite integral algebra over the field and hence a field.

Source reference. Original lemma 00GS: sources/stacks-algebra/pinned-original-source.tex, source lines 7999–7999. Navigation pointer invoked at proof line 675; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *The Brauer group of an algebraically closed field is zero.*

Proof. Let $k \subset K$ be a finite central skew field extension. For any element $x \in K$ the subring $k[x] \subset K$ is a commutative finite integral k-sub algebra, hence a field, see Algebra, Lemma 00GS. Since k is algebraically closed we conclude that $k[x] = k$. Since x was arbitrary we conclude $k = K$. □

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